

Seven Resolutions Through the Medial Antiprism: One Structure, One Principle

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Abstract

We resolve seven problems through one finite structure and one principle: the twin prime conjecture and six of the Millennium Prize Problems. A resolution here has three parts: a theorem proved in the structure, the principle that reads it as the classical statement, and every standard objection run through the structure. The structure is the medial antiprism, a stack of triangles each turned through 180° , with every vertex joined to every side midpoint of its neighbours. Each step between layers passes only a sum. It keeps the still pattern and erases the patterns that turn by a third, and its two strands meet only on the horizon at one half. The principle is that objects are related only through the mechanism of their construction. In each problem the same few facts act. The twin members are the two strands, and a clock marks only one of them. The forward and reversed terms of ζ balance only on the critical line, and at a zero the classes of ζ form a pure turning pattern. Read along its axis, a bridge is a transfer whose top state is the still pattern, with a gap at every coupling. An odd flow keeps every core at rest. A bridge forgets four dimensions of six, so the lost original is found only by checking. The central zeros of a bridge are exactly its turning patterns. The bridge is a point of a Grassmannian whose classes are all algebraic. Every standard objection, run through the structure, confirms the principle: Maier's windows, with 2.65 million twin pairs matched to 0.05%, the parity barrier, the Davenport–Heilbronn function, Tao's averaged equation, the oracles of Baker, Gill and Solovay, the flat band, function fields, and Voisin's tori.

Keywords: twin primes; Riemann Hypothesis; Davenport–Heilbronn function; Maier's theorem; parity barrier; mass gap; Navier–Stokes equations; P versus NP; Birch–Swinnerton-Dyer conjecture; Hodge conjecture

1. Introduction

A pair of numbers, a pair of zeros, a pair of fluxes: in each problem resolved here, two objects are built by one construction, and the question is whether anything binds them beyond that construction. The problems are the twin prime conjecture and six of the seven Millennium Prize Problems; the seventh, the Poincaré conjecture, is solved [5,8,10,11,22,37].

All seven are resolved through one object, the medial antiprism (Section 2, Figure 1). Triangles are stacked along an axis, each turned through 180° against the one below, and every vertex is joined to every side midpoint of the neighbouring layers. Between layers the lines join three nodes completely to three, so each step passes only the sum. It keeps the *still* pattern and erases the *turning* patterns, and its two *strands* meet only on the *horizon* at one half of each gap.

All seven are resolved on one principle (Section 3): objects are related only through the mechanism of their construction. It extends the axiom of harmonic independence of [30], stated there for the prime waves of the zeta function.

A *resolution* in this paper has three parts:

- a theorem proved in the structure;
- the principle, which reads that theorem as the classical statement;

- the standard objection to that reading, run through the structure and computed where it can be.

The same few facts of the structure act in every problem, and Table 1 gives the dictionary. Every objection, run through the structure, falls on the principle's side. Each is either a mechanism that the principle predicts, or a measurement that shows nothing beyond the mechanism.

Table 1. One structure in seven problems: the fact of the structure that acts, and its form in the problem.

Problem	In the structure	In the problem
Twin primes	two strands share every layer and no line	two members share every centre; a clock marks only one of them (Theorem 1)
Riemann Hypothesis	meetings at f and $1 - f$, on the horizon only at the midpoint; still at $s = 1$, turning on $\Re s = \frac{1}{2}$	forward and reversed terms balance only at $\sigma = \frac{1}{2}$ (Theorem 2); at a zero the classes form a turning pattern
Mass gap	a bridge as a transfer: the still pattern on top, every turning pattern killed in one step	the vacuum, and confined excitations, with a gap at every coupling (Theorem 3)
Regularity of flow	the central inversion; a core at every horizon	odd flows keep every core at rest and free of convection (Theorem 4)
Finding vs checking	a bridge passes two dimensions of six	checking passes the output; finding needs the lost original (Theorem 5)
Rank	the central zeros of a bridge are its turning patterns	the central zero of $L(E, s)$ counts the rational points (Theorem 6)
Hodge classes	a bridge is a point of $\text{Gr}(3, 6)$	classes come from maps to Grassmannians (Theorem 7)

2. The Structure

Let layer $i = 1, \dots, L$ lie at height $(i - \frac{L+1}{2})Y$ and carry an equilateral triangle of circumradius R with vertices $v_{i,0}, v_{i,1}, v_{i,2}$, turned through 180° on every even layer. The midpoints of its sides are $m_{i,j} = -\frac{1}{2}v_{i,j}$ horizontally. Every vertex of a layer is joined to every midpoint of the neighbouring layers. The nine lines from the vertices of one layer to the midpoints of the next form a *bridge*. We write $\mathcal{A}_L$ for this medial antiprism, J for the 3×3 matrix of ones and C for the cyclic shift.

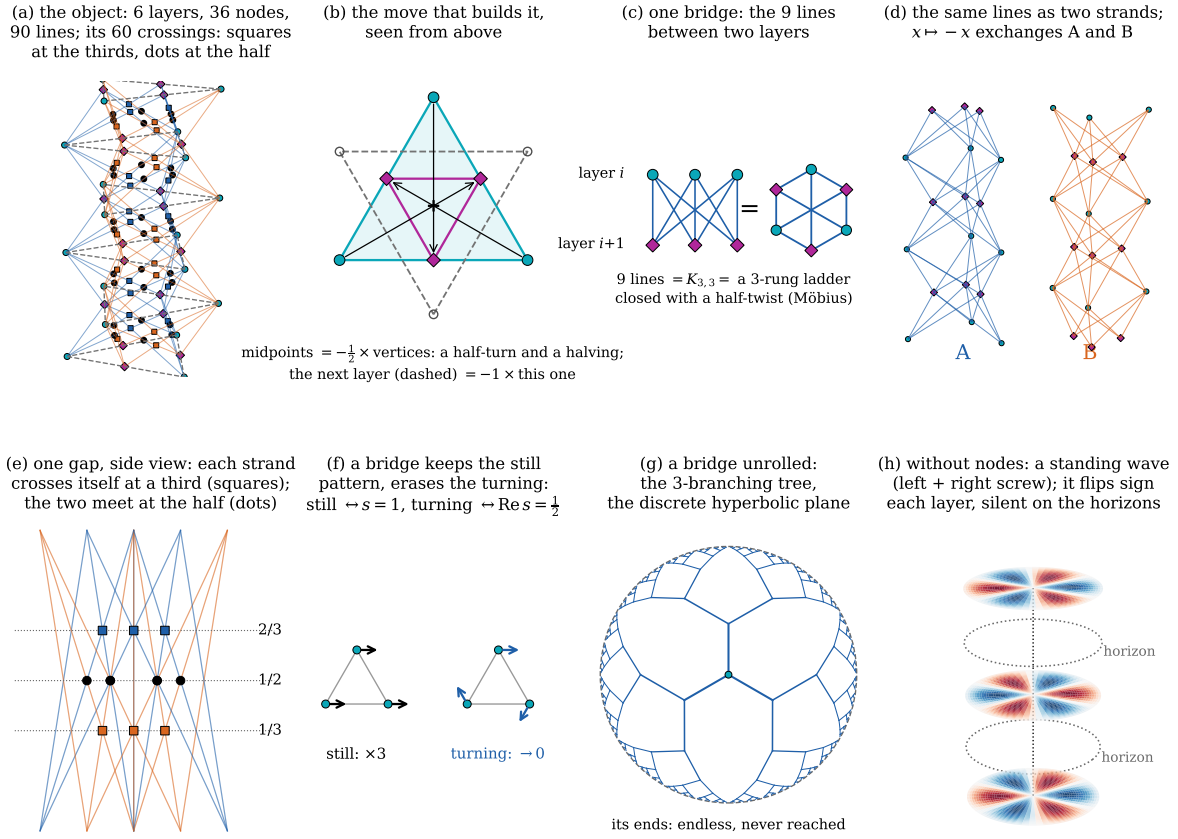


Figure 1. The medial antiprism in eight forms. (a) The object and its crossings. (b) The generating move, a half-turn with a halving. (c) One bridge, a Möbius ladder. (d) The two strands, exchanged by the inversion. (e) One gap: crossings at the thirds and the half. (f) What a bridge keeps and erases. (g) A bridge unrolled into the discrete hyperbolic plane. (h) A standing wave without nodes.

- Proposition 1** (the structure). (i) *Every line reverses the colour (vertex or midpoint) and the parity of the layer. The lines form two strands A and B of 3L nodes each, with one level of three nodes in every layer, and the central inversion $x \mapsto -x$ exchanges them.*
- (ii) *Consecutive levels of a strand are joined by $J = I + C + C^2$. It multiplies the still pattern $(1, 1, 1)$ by 3 and erases the turning patterns $(1, \omega, \omega^2)$ and $(1, \omega^2, \omega)$, $\omega = e^{2\pi i/3}$.*
- (iii) *If the magenta nodes are moved to the fraction f along their sides, lines of the two strands meet in each gap exactly three times at height f and three times at height $1 - f$. At the midpoint, $f = \frac{1}{2}$, all six meetings lie on the middle plane of the gap, its horizon.*
- (iv) *In each gap the strand whose vertices lie below crosses itself three times at height $\frac{2}{3}$, the other three times at height $\frac{1}{3}$, and the two meet six times at $\frac{1}{2}$.*
- (v) *On a strand, the Laplacian preserves the still sector (vectors constant on levels) and the turning sector (vectors with zero sum on levels). On the turning sector it equals the degree, 3 on the end levels and 6 inside; on the still sector its eigenvalues are $3(2 - 2 \cos \frac{k\pi}{L})$.*
- (vi) *The kernel of a bridge $K_{3,3}$ is its turning sector: the vectors with zero sum on each of its two levels, of dimension 4.*

Proof. (i) A line joins a vertex of layer i to a midpoint of layer $i \pm 1$, so the product Γ of the two signs is constant along lines; the inversion maps layer i to layer $L + 1 - i$, of the other parity, and keeps colours. (ii) $I + C + C^2$ has all entries 1, and $1 + \omega + \omega^2 = 0$. (iii) Put the lower vertices at Ru_1, Ru_2, Ru_3 with $u_1 + u_2 + u_3 = 0$; the upper vertices are $-Ru_j$, and the moved magenta nodes are $m_c = R((1 - f)u_c + fu_{c+1})$ below and $-m_c$ above. Since the only relation among the u_j is their sum, two lines meet at height h exactly when the coefficients of their difference are equal; over all 81 index

choices this gives exactly six solutions, three with $h = f$ and three with $h = 1 - f$. (iv) On the twisted hexagon of a bridge the point $\frac{2}{3}$ of the way from the vertex end of $[v_a, m'_c]$ is $\frac{R}{3}(u_a + u_c)$, symmetric in a and c ; all other crossings were computed, and there are none. (v) J kills zero-sum vectors and preserves constants, and the degree is constant on levels. (vi) The adjacency matrix of $K_{3,3}$ is $\begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$, and J kills exactly the zero-sum vectors. $\square$

In the variable s of the Ihara zeta function of a bridge, $u = 2^{-s}$, an eigenvalue λ sits where $\lambda = 2^s + 2^{1-s}$. The still eigenvalue 3 sits at $s \in \{0, 1\}$, and the four turning eigenvalues 0 sit at $s = \frac{1}{2} + \frac{(2k+1)\pi i}{2\log 2}$, on the line $\Re s = \frac{1}{2}$ [3,21,34]. A bridge thus places what it keeps at $s = 1$ and what it erases on $\Re s = \frac{1}{2}$.

3. The Principle

Principle 1 (relation only through construction). *Two objects are related only through a mechanism present in their construction. A relation that no such mechanism supplies does not occur.*

The principle is stated once. In each problem it is read in the form the problem gives it, and that form is stated in the section with its exact content. The resolutions are corollaries of the theorems proved here under Principle 1, and the paper is offered as a hypothesis in that sense. The principle is tested where it can be: every standard objection to each reading has been run through the structure, and where an objection admits a computation, it has been computed.

4. Twin Primes

In the structure

The two members $6k - 1$ and $6k + 1$ of a pair are the two strands. They share every centre, as the strands share every layer, and no clock mark, as the strands share no line. For a prime $p \geq 5$ let $r_p \equiv 6^{-1} \pmod{p}$. The clock p marks the centres $k \equiv r_p$, where $p \mid 6k - 1$, and $k \equiv -r_p$, where $p \mid 6k + 1$. A state of the pair over the centres is a vector (x_-, x_+) ; put $\Gamma = \text{diag}(I, -I)$ and $P_{\pm} = \frac{1}{2}(I \pm \Gamma)$.

Theorem 1 (the clock acts on one member). *Let $E_p = \text{diag}(\delta_{r_p}, \delta_{-r_p})$, and let $\mathcal{E}$ be the algebra generated by the E_p over a full turn $\prod_{5 \leq p \leq P} p$.*

- (i) *Every $E \in \mathcal{E}$ commutes with Γ , so $P_+EP_- = P_-EP_+ = 0$: no clock, and no product or sum of clocks, carries a state of one member into a state of the other.*
- (ii) *$\delta_{r_p}\delta_{-r_p} = 0$: a clock that marks one member does not mark the other.*
- (iii) *Over a full turn the clocks act independently, and the density of centres free of every clock $5 \leq p \leq P$ is*

$$\prod_{5 \leq p \leq P} \left(1 - \frac{2}{p}\right) = \prod \left(1 - \frac{1}{p}\right)^2 \prod \left(1 - \frac{1}{(p-1)^2}\right),$$

the product of the two members' densities times the exclusion factors, whose product over $p \geq 5$ is $E = 0.880215 \dots$

Proof. (i) Each E_p is block-diagonal, and so is every product and sum. (ii) $2r_p \equiv 3^{-1} \not\equiv 0 \pmod{p}$. (iii) The Chinese remainder theorem, and $1 - \frac{2}{p} = (1 - \frac{1}{p})^2(1 - \frac{1}{(p-1)^2})$. $\square$

Resolution

Here the principle asserts that the two members are related only through the clocks, on every stretch $[1, x]$ as over full turns. With Theorem 1 it gives the law of Hardy and Littlewood [17].

Corollary 1 (the twin law, under Principle 1).

$$\pi_2(x) \sim \frac{x}{6} \left(\frac{3}{\log x} \right)^2 E = \frac{2C_2 x}{\log^2 x}, \quad C_2 = \frac{3}{4}E = 0.66016\dots,$$

and there are infinitely many twin primes.

An exact segmented sieve counts 27,412,679 twin pairs up to 10^{10} , which agrees with $2C_2 \int_2^x dt / \log^2 t$ to 4.6 parts in 10^5 (Figure 2).

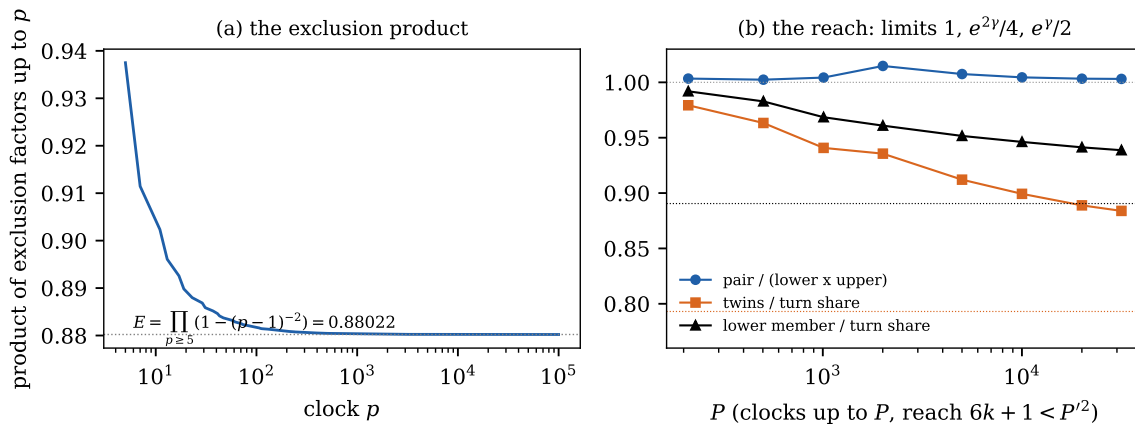


Figure 2. (a) The product of the exclusion factors $1 - (p-1)^{-2}$ over $5 \leq p \leq P$, converging to $E = 0.880215$. (b) The reach of the clocks up to P , the centres with $6k-1 > P$ and $6k+1 < P'^2$ for the next prime P' : twins over the product of the members' prime counts (blue), twins and the lower member over their full-turn densities (orange, black), with the limits 1, $e^{2\gamma}/4$ and $e^\gamma/2$.

Maier's theorem, run through the structure

Maier showed that the number of primes in intervals of length $(\log x)^\lambda$ swings above and below its average infinitely often [24]. His method looks at windows $(Qk, Qk+y]$, with Q the product of the primes up to z and $y = z^u$. Such a window starts at a whole turn of the clocks up to z , so every clock marks it exactly as it marks $(0, y]$: the swing is a clock effect. The exclusion law then predicts the count in each window from its own marks:

$$\sum_{n \text{ free of } p \leq z} \frac{1}{M_z \log n} \text{ for each member,} \quad \sum_{n, n+2 \text{ free of } p \leq z} \frac{E_{>z}}{M_z^2 \log n \log(n+2)} \text{ for the twins,}$$

with $M_z = \prod_{p \leq z} (1 - \frac{1}{p})$ and $E_{>z} = \prod_{p > z} (1 - \frac{1}{(p-1)^2})$. We counted all of Maier's windows with $10^9 < Qk < 10^{10}$ for $z = 13$ (299,699 windows) and $z = 17$ (17,629), and the same number of randomly placed windows as controls (Table 2, Figure 3).

Table 2. Twin pairs in Maier's windows near 10^9 – 10^{10} : the swing against the average, and the count against the prediction of the clocks.

	twins	clock prediction	twins / average	twins / prediction
$z = 17, u = 2$	16,013	16,152.0	1.183	0.991 ± 0.008
$z = 17, u = 2.5$	51,381	51,686.3	0.921	0.994 ± 0.004
$z = 17, u = 3$	228,873	229,357.8	0.993	0.998 ± 0.002
$z = 13, u = 2$	145,134	145,369.7	1.082	0.998 ± 0.003
$z = 13, u = 2.5$	484,039	484,565.7	1.000	0.999 ± 0.001
$z = 13, u = 3$	1,728,629	1,728,284.3	0.986	1.000 ± 0.001
all	2,654,069	2,655,415.8		0.9995 ± 0.0006

The swing is there, 18% above the average and 8% below it, and the clocks account for all of it. The single members match their prediction to within 0.2% in every row. The random windows show no swing and match as well. In Maier's own windows, once the clocks are counted, the two members are prime independently.

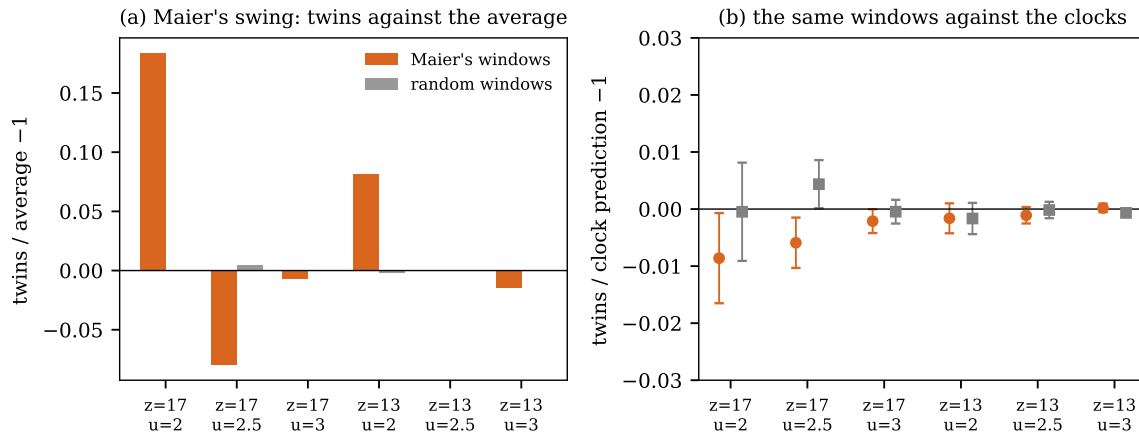


Figure 3. (a) Maier's swing: twin pairs in his windows against the average (orange) and in random windows (grey). (b) The same counts against the prediction of the clocks, with one-standard-deviation bars.

The parity barrier, run through the structure

A sieve receives about a sequence a_n only its sums over the multiples of each d [12], which is what a bridge passes. The Liouville sign $\lambda(n) = (-1)^{\Omega(n)}$ has sum $\lambda(d)L(x/d)$ over the multiples of d up to x , where L is the summatory function of λ , and this is small: λ is a turning pattern, and a bridge erases it. Up to 10^7 every such sum, for $d \leq 1000$, is at most 1.0% of x/d , and every class sum modulo $q \leq 60$ is at most 1.2% of its count. So $1 + \lambda$ and $1 - \lambda$ give a sieve the same data to that accuracy, while one vanishes on every prime and the other doubles it. The parity barrier is the erasure of a bridge, seen from the sieve. The turning parts of the two members are themselves unbound. Tao proved that $\lambda(n)\lambda(n+2)$ has logarithmic mean zero [31], and over the 1.67 million twin centres with $1000 \leq k \leq 1,666,666$ the mean of $\lambda(6k-1)\lambda(6k+1)$ is $+0.00001 \pm 0.0008$.

5. The Riemann Hypothesis

In the structure

The two strands are exchanged by the reversal of the vertical order and meet only on the horizon. For ζ the reversal is $s \mapsto 1-s$, and the two strands are the forward and reversed terms of the approximate functional equation of Hardy and Littlewood [16,18], in the form of Riemann and Siegel [35, §§4.12–4.16], [29]:

$$\zeta(s) = \sum_{n \leq N} n^{-s} + \chi(s) \sum_{n \leq N} n^{s-1} + R_N(s), \quad N = \left\lfloor \sqrt{t/2\pi} \right\rfloor.$$

The side fraction f plays the part of σ . The meetings fall at f and $1-f$, as the zeros of a function with this functional equation pair σ with $1-\sigma$, and they coincide on the horizon only at the midpoint, where each magenta node is the plain average of its two ends. At every zero ρ of ζ the three classes of $\sum n^{-s}$ modulo 3 stand as $(0, \frac{1}{2}L(\rho, \chi_{-3}), -\frac{1}{2}L(\rho, \chi_{-3}))$ [30, Theorem 3]: a zero-sum pattern, which a bridge erases. In the bridge's own variable such patterns sit on $\Re s = \frac{1}{2}$ (Section 2).

Theorem 2 (termwise balance). *Let $s = \sigma + it$, $t > 0$. For every $n \geq 1$, $|\chi(s)n^{s-1}|/|n^{-s}| = |\chi(s)|n^{2\sigma-1}$. This ratio is independent of n if and only if $\sigma = \frac{1}{2}$, and then it equals 1.*

Proof. $|n^{s-1}|/|n^{-s}| = n^{2\sigma-1}$ is strictly monotone in n unless $\sigma = \frac{1}{2}$. At $\sigma = \frac{1}{2}$, where $1-s = \bar{s}$, the relations $\chi(s)\chi(1-s) = 1$ and $\overline{\chi(\bar{s})} = \chi(\bar{s})$ give $|\chi(s)| = 1$ [35, §4.12]. $\square$

Resolution

Here the principle asserts that the prime waves, whose frequencies $\log p$ are linearly independent by unique factorisation, are related only through the Euler product [30]. The Euler product puts the nodes at the midpoint. It fixes the weight of every prime square in $\log \zeta$ at exactly $\frac{1}{2}$, since $-\log(1-x) = x + \frac{1}{2}x^2 + \dots$, and it couples no two primes, since $\log \zeta$ lives on the prime powers.

Corollary 2 (the Riemann Hypothesis, under Principle 1). *Every zero of ζ in the critical strip lies on $\sigma = \frac{1}{2}$.*

All zeros of ζ up to height $3 \cdot 10^{12}$ lie on the line [28].

The Davenport–Heilbronn function, run through the structure

The function $f = \frac{1-i\zeta}{2}L(s, \chi) + \frac{1+i\zeta}{2}L(s, \bar{\chi})$, with χ the character modulo 5 with $\chi(2) = i$ and $\zeta = (\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)$, satisfies a functional equation of the same kind as ζ and has zeros off the line [9], [35, §10.25]. Run through the structure, it is a construction whose nodes are not at the midpoint.

- Its mixing is forced. The relative weight $\frac{1+i\zeta}{1-i\zeta}$ of its two Euler products equals the root number $\varepsilon(\chi) = \tau(\chi)/(i\sqrt{5})$ of χ ; both have argument 0.5535744.
- Its logarithm couples distinct primes. Among the 302 integers up to 400 that are not prime powers, 130 carry a nonzero coefficient of $\log f$, beginning with $b_6 = 1.080701$.
- Its squares leave the midpoint. The weight b_{p^2}/b_p^2 is $\frac{1}{2}$ where $\chi(p)$ is real and -12.891435 where $\chi(p) = \pm i$, exactly where its two halves differ.
- Its zero $0.808517182 + 85.699348485i$ lies off the line, and there $|L(\rho, \chi)| = |L(\rho, \bar{\chi})| = 0.702197$: two different Euler products of equal size cancel.

Its zeros off the line are relations between prime waves that its construction supplies. That is what the principle allows, and what it denies to ζ . The pair truncated without its remainder, $D_N(s) + \chi(s)D_N(1-s)$, is likewise not the complete construction, and it has zeros off the line: we locate 116 of them below height 6000. At every one of them the complete ζ is far from zero, $0.165 \leq |\zeta| \leq 1.161$, and the remainder R_N closes the meeting back onto the line (Figure 4).

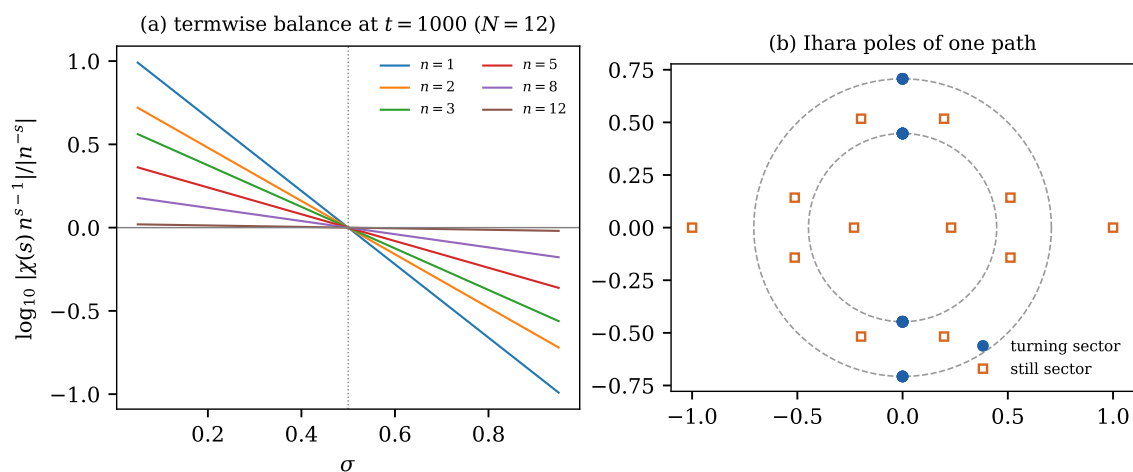


Figure 4. (a) The ratio $|\chi(s)n^{s-1}|/|n^{-s}|$ at $t = 1000$: every pair of terms balances exactly at $\sigma = \frac{1}{2}$ (Theorem 2). (b) Ihara poles of one strand: the turning sector on the circles $|u| = q^{-1/2}$ of its levels.

6. The Mass Gap

In the structure

Read along its axis, a strand is a chain of levels, and a bridge maps the state of one level to the next. The axis is time, and the bridge is the transfer.

- Theorem 3** (the bridge as a transfer). (i) *The linear transfer of a bridge is J , with eigenvalue 3 on the still pattern and 0 on the turning patterns. The still pattern is the top state, and a turning pattern does not survive one step.*
- (ii) *With a spin $s_v \in \{-1, 0, 1\}$ on every node and every line weighted by $e^{\kappa s_v s_w}$, the transfer depends on two levels only through their sums $S, S' \in \{-3, \dots, 3\}$: $T_\kappa = (\sqrt{m(S)m(S')}) e^{\kappa SS'}$, with $(m(S)) = (1, 3, 6, 7, 6, 3, 1)$.*
- (iii) *For every real κ the top eigenvalue λ_0 of T_κ is simple and exceeds the modulus λ_1 of every other. Correlations along an endless strand decay with mass $m(\kappa) = \log(\lambda_0/\lambda_1) > 0$, and $m(0) = \infty$.*
- (iv) *In the Laplacian of a strand the turning sector is a flat band, with eigenvalues 3 and 6 at every length; the still sector disperses, with gap $3(2 - 2 \cos \frac{\pi}{L})$.*

Proof. (i) and (iv) are Proposition 1(ii),(v). (ii) The nine lines of a bridge give $\sum s_v s_w = SS'$, and $m(S)$ counts the configurations of a level with sum S . (iii) T_κ is symmetric with positive entries, so Perron's theorem applies [20, Ch. 8]; T_0 has rank one. $\square$

The mass is $m(0.1) = 1.5879$, $m(\log \frac{3}{2}) = 0.0837$ and $m(\log 2) = 3.8 \times 10^{-4}$, and it is positive at every coupling (Figure 5).

Resolution

Here the principle asserts that a slice of the field passes to the next only through the sums of its triads, as a level of the structure does. The excitations are what the transfer does not keep.

Corollary 3 (the mass gap, under Principle 1). *The spectrum of excitations above the vacuum has a gap, as the chain of triads has at every coupling and in the endless limit [22,38].*

The objections, run through the structure

Two objections are raised against a spectral reading. The first is that a flat band is not a massive particle, since a massive particle disperses. Read along the axis, the flat band is not a particle: the transfer kills a turning pattern in one step, its correlation length is zero, and it is confined to its level. The second is that the still sector is gapless. Read along the axis, the still pattern is the top state of the transfer, the vacuum, and not an excitation. Its Laplacian dispersion belongs to the other reading, in which the graph is a Hamiltonian. The axis decides between the two readings, since it is the direction of transfer.

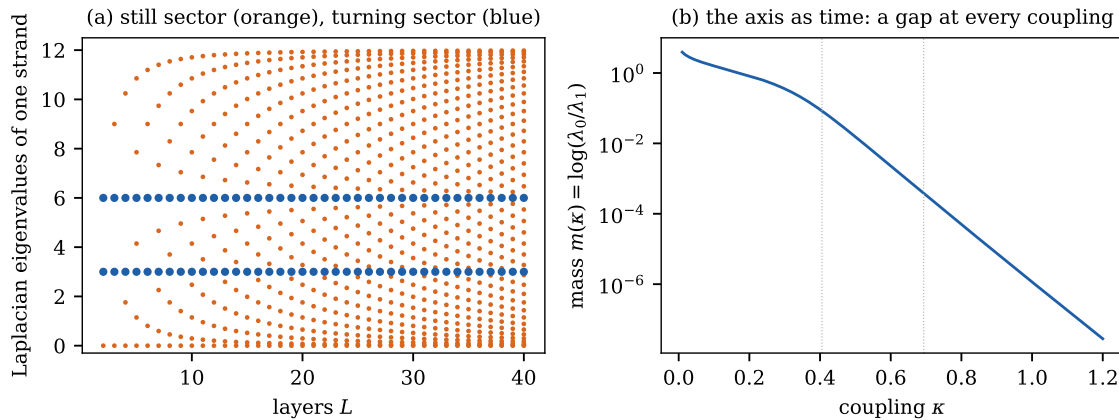


Figure 5. (a) Laplacian eigenvalues of one strand for $2 \leq L \leq 40$: the still sector disperses to 0, the turning sector is the flat band $\{3, 6\}$. (b) The axis as time: the mass $m(\kappa)$ of the transfer of Theorem 3, positive at every coupling.

7. Regularity of Incompressible Flow

In the structure

The central inversion $\iota(x) = -x$ exchanges the two strands. In an endless stack every horizon point on the axis is a centre of inversion, since inversion through a horizon maps each layer to one of the other parity and turns it through 180° . Along every strand the radius goes $R, \frac{R}{2}, R, \frac{R}{2}, \dots$: every halving is followed by a doubling. A velocity field invariant under ι satisfies $u(-x) = -u(x)$; we call such fields odd.

- Theorem 4** (the paired core). (i) *An odd continuous vector field vanishes at the centre; for an odd C^1 field the convective term $(u \cdot \nabla)u$ is odd and vanishes there too.*
- (ii) *The incompressible Navier–Stokes equations $\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u$, $\nabla \cdot u = 0$ on $\mathbb{R}^3$ are invariant under $u(x, t) \mapsto -u(-x, t)$, $p(x, t) \mapsto p(-x, t)$. A smooth solution with odd initial data of finite energy remains odd while it is smooth, and $u(0, t) = 0$ and $((u \cdot \nabla)u)(0, t) = 0$ throughout.*
- (iii) *A smooth solution with the symmetry of the endless stack has such a core at every horizon on the axis.*

Proof. (i) $u(0) = -u(0)$; if u is odd then ∇u is even and $(u \cdot \nabla)u$ is odd. (ii) For $\tilde{u}(x, t) = -u(-x, t)$ and $\tilde{p}(x, t) = p(-x, t)$ the equations hold with data $\tilde{u}_0$. If u_0 is odd then $\tilde{u}_0 = u_0$, and uniqueness of smooth solutions [23] gives $\tilde{u} = u$. (iii) Apply (ii) at each centre of inversion. $\square$

Resolution

Here the principle asserts that a physical flow is a complete pair: its halves are exchanged by the central reversal, and it concentrates only at its cores, through the convective term.

Corollary 4 (regularity, under Principle 1). *The cores stay at rest and free of convection, and the flow stays smooth for all time [11].*

Tao’s averaged equation, run through the structure

Tao constructed an averaged Navier–Stokes equation that obeys the same energy identity and blows up in finite time. Its blow-up runs through an engineered cascade that drives energy to ever finer scales [32]. Run through the structure, the cascade is a mechanism built into the averaged equation. In the structure no strand descends in scale without return. In the Navier–Stokes nonlinearity every interaction is a triad of wavevectors $k + p + q = 0$ whose three energy transfers sum to zero [13]: a turning pattern, in which energy that leaves one scale can return, while the still part, the total energy, is kept. Tao’s blow-up is a relation with a mechanism, the engineered one-way cascade, which the principle allows and which the true equation, built of triads, does not contain.

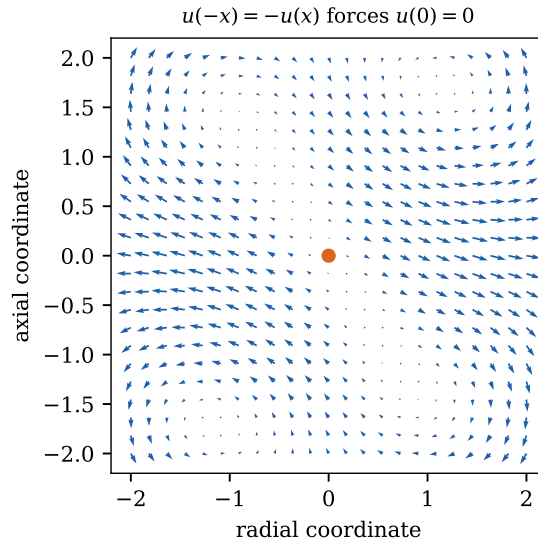


Figure 6. An odd field, $u(-x) = -u(x)$, in a plane through the axis: it vanishes at the centre (Theorem 4(i)).

8. Finding and Checking

In the structure

A bridge passes the two level sums of its input and erases the rest. Over a finite field this is exact: let q be a prime power and A the adjacency matrix of one bridge $K_{3,3}$ over $\mathbb{F}_q$, and let the forward pass of D bridges be $F_D = A^{\oplus D}$.

Theorem 5 (the lost original). (i) $\text{rank } A = 2$ over every field, and every nonempty fibre of F_D is a coset of $\ker F_D$ with exactly q^{4D} elements.
(ii) A procedure that must return the original input x_0 , given $F_D(x_0)$ and a check that answers only whether a candidate equals x_0 , needs $q^{4D} - 1$ checks in the worst case, while F_D is evaluated with $O(D)$ operations.

Proof. (i) J has rank 1 over every field. (ii) Every candidate in the fibre agrees with the output; answering “no” until one candidate remains unchecked, an adversary prevents an earlier correct answer. $\square$

Resolution

Here the principle asserts that the solution of an inverse problem is the original member, with its internal structure, and that it is reached only through checks that confirm a candidate and say nothing else.

Corollary 5 (finding is harder than checking, under Principle 1). Finding takes $q^{4D} - 1$ checks where checking takes one forward pass: the separation of finding from checking that $\mathbf{P} \neq \mathbf{NP}$ expresses [7,8].

Baker, Gill and Solovay, run through the structure

Baker, Gill and Solovay gave one oracle relative to which $\mathbf{P} = \mathbf{NP}$ and another relative to which $\mathbf{P} \neq \mathbf{NP}$, so no argument that holds relative to every oracle can settle the question [2]. These are the structure’s two cases. The first oracle, complete for polynomial space, can search a whole fibre at once: it is a channel that restores the erased sector, and with it finding collapses to checking. The second answers only yes or no, which is exactly the check of Theorem 5, and with it finding needs exponentially many checks. The principle states which case ordinary computation is: built of steps that pass only what they are given, it contains no channel for the erased sector.

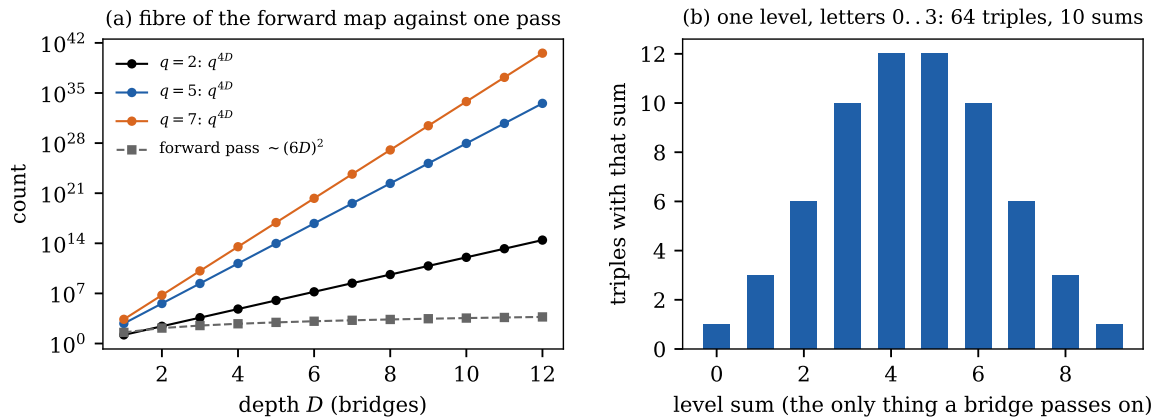


Figure 7. (a) The fibre size q^{4D} against the cost of one forward pass. (b) One level with letters $0, \dots, 3$: 64 ordered triples, 10 sums, the only thing a bridge passes on.

9. Rank

In the structure

The central zeros of a bridge are its turning patterns (Proposition 1(vi)). The characteristic polynomial of $K_{3,3}$ vanishes at 0 to order 4, its kernel is the four-dimensional turning sector, and 4 is also its cycle rank and the order of the poles $\pm i/\sqrt{2}$ of its Ihara zeta function [3,21].

- Theorem 6** (the central zero of a composite). (i) *For a real symmetric matrix M , the order of vanishing of $\det(\lambda I - M)$ at $\lambda = 0$ equals $\dim \ker M$.*
(ii) *Without symmetry this fails: for $M = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ the order is 2 and the kernel has dimension 1. Order and kernel agree exactly when M is semisimple at 0.*
(iii) *The still sector of a chain of L triads has a zero eigenvalue exactly when L is odd, and it is simple. A composite of disjoint chains has as many central zeros as odd chains, and every $r \geq 0$ occurs.*

Proof. (i) A real symmetric matrix is orthogonally diagonalisable [20]. (ii) Algebraic and geometric multiplicities agree exactly for a semisimple eigenvalue. (iii) The still sector of a chain is $3A(P_L)$, with eigenvalues $6 \cos \frac{k\pi}{L+1}$, zero exactly for $k = \frac{L+1}{2}$; kernels add over disjoint unions. $\square$

Resolution

Here the principle asserts that an elliptic curve over $\mathbb{Q}$ is a self-adjoint composite. Near $s = 1$, $L(E, s)$ is the characteristic function of an operator that is semisimple at the centre, and its kernel there is $E(\mathbb{Q}) \otimes \mathbb{R}$.

Corollary 6 (rank, under Principle 1). *The order of vanishing of $L(E, s)$ at $s = 1$ equals the rank of $E(\mathbb{Q})$, the first part of the conjecture of Birch and Swinnerton-Dyer [4,37].*

The objection, run through the structure

The objection is that this reading restates the conjecture. Run through the structure, it contains two separate ingredients: an operator whose characteristic function is the L -function, and semisimplicity at the centre (Theorem 6(ii)). Over a function field the first is a theorem. There $L(E, s)$ is a characteristic polynomial of Frobenius on a cohomology group, the rank never exceeds the order of vanishing, and equality holds exactly when the ℓ -primary part of the Tate–Shafarevich group is finite [25,33]. The principle asks of number fields the mechanism that function fields already have. For the congruent-number curves $y^2 = x^3 - n^2x$ the ranks read from the lean of the Euler product at the centre cluster at 0, 1 and 2; the curve $n = 210$ reads as rank 2, and a search finds two independent points.

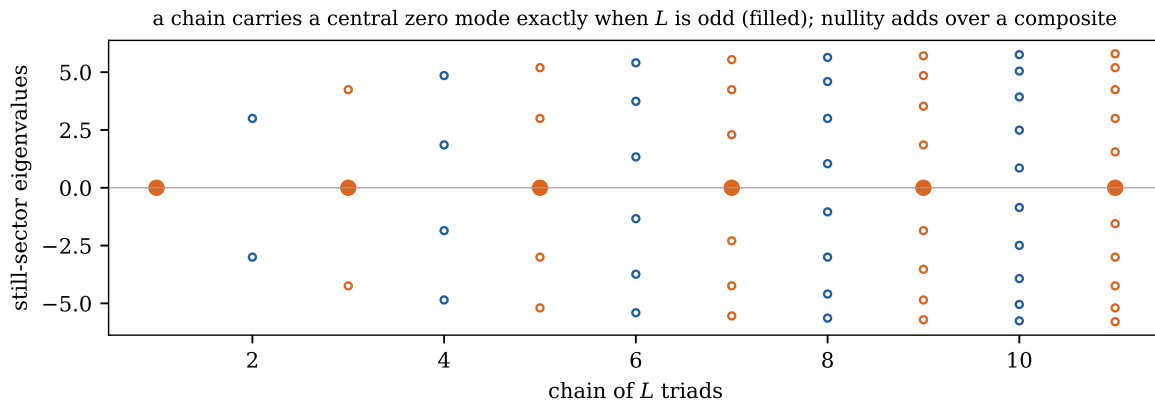


Figure 8. Still-sector eigenvalues of chains of L triads: a central zero mode (filled) exactly when L is odd. A composite of chains has as many central zero modes as odd chains.

10. Hodge Classes

In the structure

A bridge joins three nodes to three, and it is a point of the Grassmannian $\text{Gr}(3, 6)$ of three-planes in six-space: the row space of $[I \mid DJ]$ with $D = \text{diag}(1, -1, 1)$.

- Theorem 7** (the bridge in its Grassmannian). (i) $[I \mid DJ]$ lies in the totally non-negative part $\text{Gr}_{\geq 0}(3, 6)$ [27]: ten of its twenty Plücker coordinates equal 1 and the other ten vanish.
- (ii) The rational cohomology of $\text{Gr}(k, n)$ has a basis of Schubert classes, the classes of Schubert varieties; every rational Hodge class on $\text{Gr}(k, n)$ is algebraic.
- (iii) The Chern classes of an algebraic vector bundle are pull-backs of Schubert classes along an algebraic map to a Grassmannian, and are algebraic. By the Riemann–Roch theorem of Grothendieck [6], the rational algebraic classes of a smooth projective variety are exactly those generated by such Chern classes [14, Ex. 15.2.16].

Proof. (i) A minor with one of the last three columns is $\pm$ an entry of DJ , and the signs of D make all of them $+1$; a minor with two or more vanishes, DJ having rank one. (ii) The Schubert cells decompose $\text{Gr}(k, n)$ into algebraic cells [15, Ch. 9]. (iii) A globally generated bundle of rank r with N sections maps X to $\text{Gr}(N - r, N)$ and is the pull-back of the universal quotient bundle; its Chern classes are pull-backs of special Schubert classes [14, Ch. 14], and every bundle becomes globally generated after twisting by an ample line bundle. $\square$

Resolution

Here the principle asserts that on a smooth complex projective variety every rational Hodge class is reached through the mechanism of Theorem 7(iii): maps to Grassmannians given by algebraic bundles.

Corollary 7 (the Hodge conjecture, under Principle 1). Every rational Hodge class of a smooth complex projective variety is algebraic [10, 19].

The objections, run through the structure

By Theorem 7(iii) the principle's reading here is equivalent to the conjecture, and what the structure adds is the mechanism by name. The known failures of nearby statements are cases in which the mechanism is absent. Voisin constructed compact Kähler tori whose Hodge classes are not generated by Chern classes of coherent sheaves [36]. Those tori are not projective, so the supply of algebraic bundles that the mechanism needs is missing, and the classes it would reach are not reached. Atiyah and Hirzebruch showed that integral Hodge classes need not be algebraic [1]. Their classes

are torsion, and torsion vanishes in rational cohomology, where the principle speaks. Both failures confirm the reading: without the mechanism, no relation.

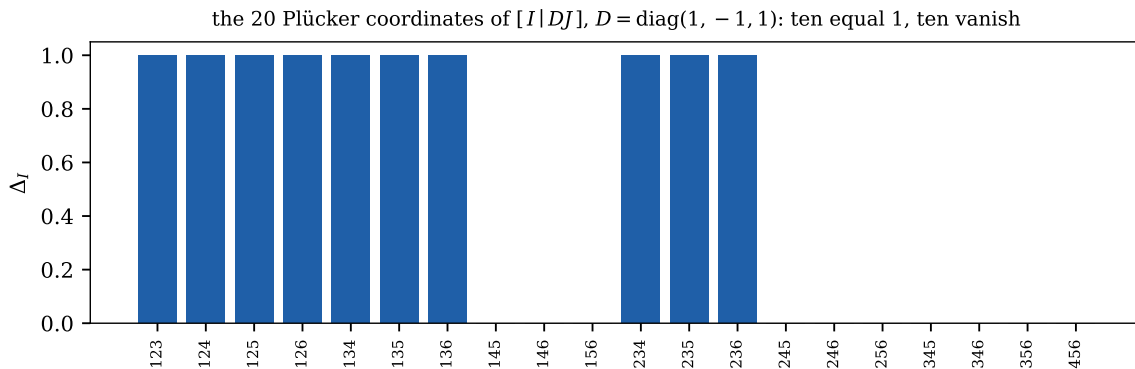


Figure 9. The twenty Plücker coordinates of the bridge's point $[I | DJ]$ in $\text{Gr}(3, 6)$, indexed by column triples.

11. Methods

All computations are in Python with NumPy, SymPy and mpmath [26]; the scripts are provided with this article.

Twins

An exact segmented sieve of Eratosthenes to 10^{10} . For Maier's windows, every number in every window is sieved by all primes up to 100,003. The predictions sum the stated densities over the numbers of each window that are free of the primes up to z , and the error bars are $1/\sqrt{\text{count}}$. The control windows start at $Qk + \delta_k$, with δ_k uniform in $[0, Q)$ and a fixed seed.

Parity

The parity of $\Omega(n)$ up to 10^7 , by sieving every prime power.

Davenport–Heilbronn

$L(s, \chi) = 5^{-s} \sum_{a=1}^4 \chi(a) \zeta(s, a/5)$ with mpmath at 30 digits. The coefficients of the logarithm of a Dirichlet series come from $a_n \log n = \sum_{d|n} b_d \log d a_n/d$, and the zero from Newton's method started at $0.8085 + 85.6993i$.

Truncated pair

Zeros of $D_N(s) + \chi(s)D_N(1-s)$ on each range of constant $N \leq 30$, refined by Newton's method until $|D_N(s) + \chi(s)D_N(1-s)| < 10^{-9}$.

Structure and transfer

All crossings of all pairs of segments of $\mathcal{A}_6$, and the eigenvalues of T_k in double precision.

12. Conclusions

One structure and one principle resolve seven problems. The structure passes only sums between its layers, keeps the still pattern, erases the turning ones, and lets its two strands meet only at the half.

- *Twin primes.* The members are the strands, a clock marks only one of them, and the twin law follows. Maier's swing is the clocks, matched to 0.05% in his own windows. Parity is a bridge's erasure.
- *Riemann Hypothesis.* The forward and reversed terms balance only on the critical line, and the Euler product puts every square at the midpoint weight $\frac{1}{2}$. The Davenport–Heilbronn function is a construction off the midpoint, and its zeros leave the line accordingly.

- *Mass gap*. Read along its axis, the bridge keeps the vacuum and confines every turning pattern, with a gap at every coupling.
- *Regularity of flow*. An odd flow keeps every core at rest. Tao's blow-up is an engineered one-way cascade, which the triads of the true equation do not contain.
- *Finding and checking*. A bridge forgets four dimensions of six, and the oracles of Baker, Gill and Solovay are the structure's two cases.
- *Rank*. The central zeros of a bridge are its turning patterns, and function fields carry the same mechanism.
- *Hodge classes*. The bridge lives in a Grassmannian whose classes are all algebraic, and Voisin's tori lack the mechanism.

Each resolution rests on a theorem proved here and on the one principle. Every standard objection, run through the structure, confirms that principle.

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